

EF5342 Financial Systems, Markets and Instruments

Week 4 — The Money Markets

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We review the money markets and the securities traded there, and discuss why the money markets are important in our financial system. Topics:

- The money markets defined
- The purpose of money markets
- Money market instruments
- Money market yields
- Money market funds

The Money Markets Defined

The term “money market” is a **misnomer**: money (currency) is not actually traded there. The securities are **short term with high liquidity and low risk** — therefore close to being money.

- **Short maturity**: typically overnight to one year (often 1–3 months or 90 days)
- **Low risk**: high-quality instruments with minimal credit and interest rate risk — often backed by governments, top-rated banks, or strong corporations
- **High liquidity**: assets can usually be sold or redeemed quickly with little price impact
- **Purpose**: cash management rather than long-term investment or capital appreciation — returns are modest compared to stocks or bonds

Three core economic functions: **cash management** (funding day-to-day liquidity needs), **monetary policy transmission** (the Fed uses money market rates as its primary policy lever), and **safe asset provision** (capital preservation in near-zero-default-risk instruments).

Money Market vs. Capital Market — and the Participant Ecosystem

Aspect	Money Market	Capital Market
Maturity	Short-term (< 1 year)	Long-term (> 1 year)
Risk/Return	Low risk, low return	Higher risk, higher potential return
Instruments	Debt only (mostly)	Stocks + long-term bonds
Purpose	Liquidity & cash management	Long-term financing & investment

The market operates across a layered ecosystem:

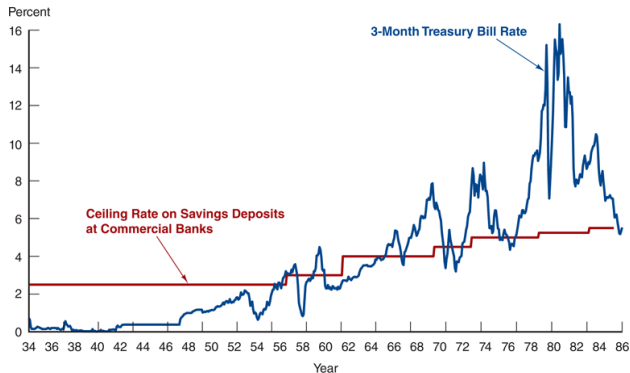
- **Cash suppliers (lenders):** MMFs, corporations, GSEs, foreign central banks, retail investors
- **Cash demanders (borrowers):** primary dealers, commercial banks, the US Treasury (T-bill issuance), corporations issuing CP
- **Central intermediary:** the Federal Reserve sets the policy rate corridor anchoring all short-term rates
- **Regulators:** SEC governs MMFs under Rule 2a-7; Fed and OCC supervise banks; FINRA oversees dealer activity

Why Do We Need Money Markets?

The banking industry *should* handle short-term investment and financing — banks have an **information advantage** — but banks are **heavily regulated**, creating a distinct cost advantage for money markets over banks.

- **Reserve requirements** create additional costs for banks that money markets do not bear
- **Regulation Q** (Banking Act of 1933) had two main components: a prohibition on paying any interest on demand deposits (checking accounts), and maximum interest rate ceilings on time and savings deposits at commercial banks (later extended to thrifts like savings & loans)
- The limits were not relevant until the 1950s; in the decades that followed, the problem became apparent

3-Month T-Bill Rates and Interest Rate Ceilings



Source: <http://www.stlouisfed.org/default.cfm>.

Figure: Three-month Treasury bill rate and ceiling rate on savings deposits at commercial banks, 1933–1986

Money Market Instruments

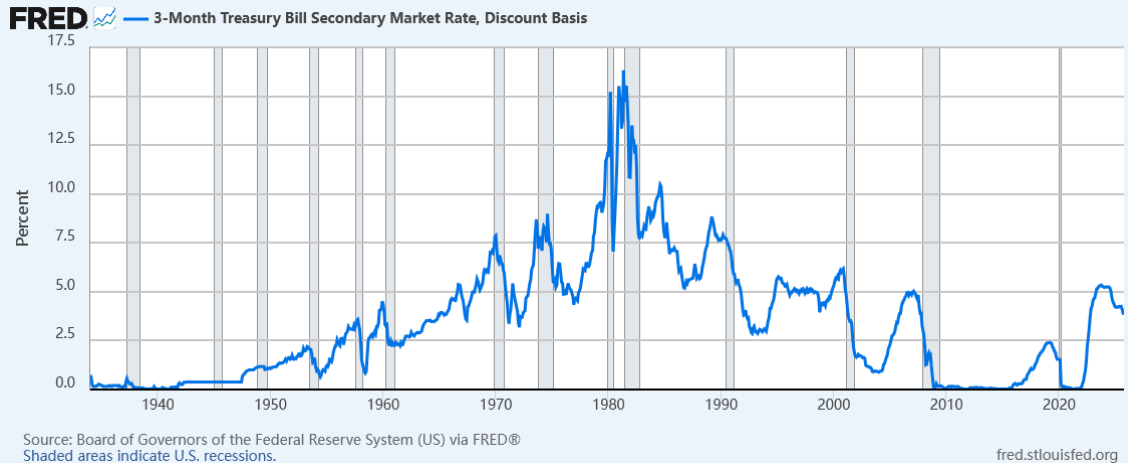
The six core instruments:

- Treasury bills (T-bills)
- Federal funds (fed funds)
- Repurchase agreements (repos or RP)
- Commercial paper (CP)
- Negotiable certificates of deposit (CD)
- Banker's acceptances (BA)

Treasury bills (T-bills)

- Short-term debt obligations of the U.S. Treasury
- Virtually default-risk free, highly liquid, little interest rate risk — considered the only risk-free asset in markets
- The Federal Reserve buys and sells T-bills to implement monetary policy
- Strong international demand as a safe-haven investment

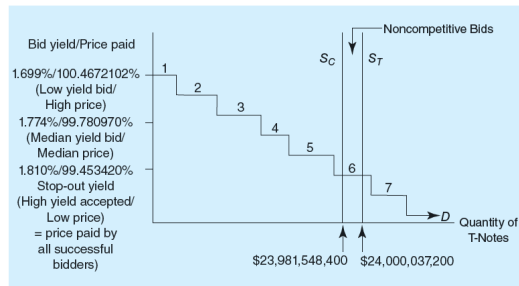
3-Month T-Bill Rate



T-Bill Auctions in the Primary Market

- Maturities of 4-, 8-, 13-, 17-, 26-, and 52-week are auctioned on varying weekly schedules
- Bids are submitted by government securities dealers, financial and nonfinancial corporations, and individuals
- Bids are either **competitive** (specifying the discount rate accepted) or **noncompetitive** (accepting the clearing rate)
- All bidders pay the lowest accepted price (highest accepted yield — the *stop-out yield*); bidders at the stop-out yield may receive a pro-rata share
- Most secondary trading occurs directly through brokers and dealers

Figure 6-7 Treasury Auction Results



The Secondary Market for T-Bills

The secondary market for T-bills is the **largest of any U.S. money market instrument**.

- Primary dealers “make” a market in T-bills by buying the majority sold at auction and by creating an active secondary market
- Primary dealers trade for themselves and for customers
- T-bill purchases and sales are book-entry transactions conducted over Fedwire

How to Calculate T-Bill Prices: Discount Yield Quotes

T-bills are quoted using the **discount yield**. There are no coupons — the price you pay is a discount from par value, and the difference implies the “interest” earned over the investment period. The **ask** price is what you pay to buy from a dealer; the **bid** price is what you receive when selling to a dealer.

Maturity	Days to maturity	Bid (%)	Ask (%)
16-Mar-23	6	4.390	4.380
13-Apr-23	34	4.500	4.490
6-Jun-23	88	4.773	4.763
10-Aug-23	153	4.865	4.855

Source: The Wall Street Journal, 10 March 2023

T-Bill Price Calculation

Price of a T-bill (par value \$10,000, 360-day year):

$$\text{Price} = 10,000 \times \left[1 - i \times \frac{n}{360} \right], \quad n = \text{days until maturity}, \quad i = \text{prevailing yield}$$

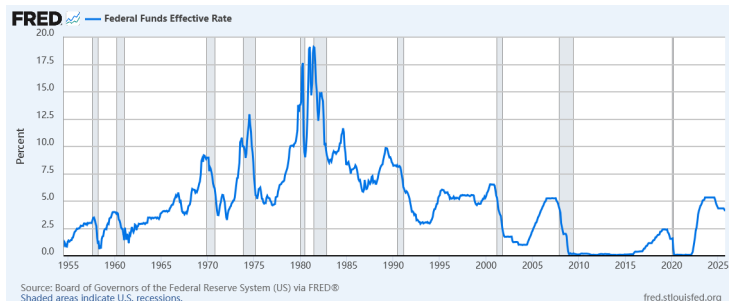
Example — T-bill maturing 10 August 2023 ($n = 153$):

- Ask price = $10,000 \times (1 - 4.855\% \times 153/360) = \$9,793.66$
- Bid price = $10,000 \times (1 - 4.865\% \times 153/360) = \$9,793.24$
- **Q:** why is the bid price lower than the ask price?

Federal Funds

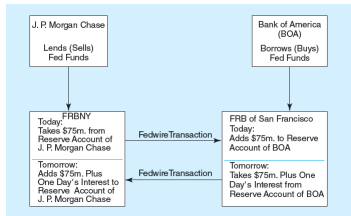
Federal funds = balances that banks hold in their reserve accounts at the Federal Reserve. The fed funds market involves **overnight, unsecured lending** of reserve balances between depository institutions: banks with excess reserves lend; banks with deficient reserves borrow. Transactions settle via **Fedwire** (the Fed's real-time gross settlement system).

- The **Effective Federal Funds Rate (EFFR)** is the volume-weighted median of these transactions; the FOMC sets a target range for EFFR as its primary policy rate



A Federal Funds Transaction

Figure 5-3 Federal Funds Transaction



Today — loan initiation:

- J.P. Morgan Chase has extra reserves; Bank of America needs more (regulatory requirements, outflows, avoiding overdrafts)
- J.P. Morgan lends (“sells”) \$75 million in fed funds to BOA — an unsecured overnight loan; BOA repays principal + interest tomorrow

Tomorrow — repayment with interest:

- BOA repays \$75m principal plus one day's interest at the agreed fed funds rate
- Via Fedwire in reverse: FRB San Francisco debits \$75m + interest from BOA; FRBNY credits \$75m + interest to J.P. Morgan
- The lender earns interest income; the borrower pays the cost of overnight reserves

Repurchase Agreements (Repos)

A **repurchase agreement (repo or RP)** is the sale of a security with an agreement to buy it back at a **higher price** in the future — essentially a short-term **collateralized** loan (typical collateral: U.S. Treasury securities).

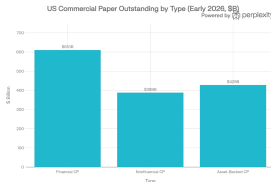
- The repo rate is typically **lower** than the Fed funds rate because repos are collateralized loans
- If collateralized by risky assets, the repo may involve a **haircut** — e.g., to borrow \$100 the repo seller must sell securities currently worth \$102 (a \$2 haircut)
- A **reverse repo** is the purchase of a security with an agreement to sell it back in the future

Commercial Paper

Commercial paper (CP) is short-term **unsecured** debt issued by corporations to fund working capital — typically accounts receivable, inventory financing, or bridge funding.

- Maturities of 1–270 days; total outstanding $\approx$ \$1.45 trillion (July 2026)
- Ratings are essential — most CP is rated A-1/P-1 (prime) or A-2/P-2 by Moody's/S&P
- Usually held to maturity; no active secondary market; yields quoted on discount yield (like T-bills)

Three main types: **nonfinancial CP** (industrial companies), **financial CP** (bank holding companies, finance subsidiaries), and **asset-backed CP** (ABCP, issued by conduits backed by receivables or other assets).



Negotiable Certificates of Deposit

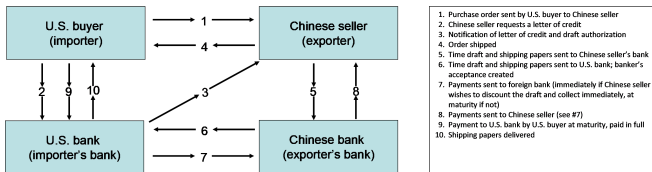
A **negotiable CD (NCD)** is a bank-issued time deposit that specifies the interest rate and maturity date.

- Denominations range from \$100,000 to \$10 million; \$1 million is the most common
- Unlike standard retail CDs, NCDs have market-determined interest rates and can be liquidated before maturity by selling to another investor
- Often purchased by money market mutual funds with pools of funds from individual investors

Banker's Acceptances

A **banker's acceptance (BA)** is a short-term debt instrument used in **international trade finance** — to facilitate secure payments between importers and exporters, especially when the parties do not fully trust each other or extended payment terms are needed.

- A **time draft** (a written order to pay a specified amount at a future date) that has been “accepted” by a bank
- By accepting, the bank **unconditionally promises** to pay the face value at maturity — replacing the importer's credit risk with the bank's (usually much stronger) credit quality



Why Banker's Acceptances Are Attractive

Party	Main benefit
Exporter	Bank's payment guarantee instead of relying on unknown importer
Importer	Deferred payment terms (90–180 days credit) without paying upfront
Both	Counterparty risk significantly reduced
Banks	Fees for acceptance; can create highly marketable short-term paper

BAs trade at a discount in the secondary money market and are among the **safest short-term instruments**:

- Often safer than commercial paper because of the bank's unconditional guarantee
- Typically yield slightly more than T-bills but less than regular CP, due to low risk
- In short: a BA transforms a risky importer's promise into a safe, bank-guaranteed, tradable short-term credit instrument

Money Market Instruments: Summary

Instrument	Issuer	Maturity	Key features
Treasury bills	Government (U.S. Treasury)	4 weeks–1 year	Zero-coupon, sold at discount, very safe
Commercial paper	Corporations	1–270 days	Unsecured promissory notes, short-term corporate borrowing
Negotiable CDs	Banks	1 month–1 year	Time deposits, often negotiable (NCDs)
Repos	Banks, dealers, institutions	Overnight–few weeks	Collateralized short-term loans (usually government securities)
Banker's acceptances	Banks (accepting trade drafts)	30–180 days	International trade, bank-guaranteed
Federal funds	Banks (interbank)	Overnight	Unsecured loans between banks to meet reserve requirements

Money Market Instruments Outstanding

Instrument	Issuer	Maturity	Key feature	Approx. size
Treasury bills	US Treasury	4–52 weeks	Zero default risk; auctioned weekly	\$7.25T (Aug 2026)
Repos	Dealers, banks	Overnight–3 months	Secured by Treasuries/MBS	\$4.4T triparty o/s (end-2025)
Commercial paper	Corporations, banks, SPVs	1–270 days	Unsecured (non-fin.) or asset-backed	\$1.45T (July 2026)
Negotiable CDs	Banks	2 weeks–1 year	Bank liability; tradeable secondary market	~\$1.7T (SIFMA, 2024)
Federal funds	Banks	Overnight	Interbank reserve lending; Fed-targeted	~\$100B/day (Aug 2026)
Eurodollar deposits	Offshore banks	Overnight–1 year	USD outside US; no Fed regulation	Multi-\$T est.; shrank post-LIBOR
Money market funds	Investment companies	Daily liquid	Pooled vehicle investing in all above	\$7.98T AUM (Sept 2026)

Sources: Treasury MSPD; OFR; FRED (DTBSPCKNMM); SIFMA; NY Fed; ICI; BIS. Figures as of the dates shown; retrieved Sept 2026.

The Money Market: Safe, but Not Risk-Free

Money market instruments are **safe, but not risk-free**. Investors require higher yields on money market securities than on T-bills:

- Spreads (risk premia) often increase during crisis periods
- Some instruments even defaulted — e.g., commercial paper issued by Lehman Brothers became worthless in 2008

International Money Markets: Eurodollars

Eurodollars are US dollar-denominated deposits held **outside the United States** — in London, Singapore, Hong Kong, the Cayman Islands, and other offshore centers. The term “Euro” is historical and geographical; any offshore USD deposit qualifies.

- Not subject to Federal Reserve reserve requirements or FDIC insurance — giving offshore banks both higher flexibility and higher credit risk
- Historically, the rate offered for borrowing Eurodollar funds was the **London Interbank Offered Rate (LIBOR)**, which ceased in 2023; offshore USD now prices mostly off SOFR-linked benchmarks
- Estimated outstanding volume: on the order of \$10 trillion in deposits and loans (order-of-magnitude estimate — the market has contracted since the LIBOR era); the linked derivatives footprint is far larger: total OTC derivatives notional was \$844.6 trillion at end-2025, of which \$669.5 trillion in interest-rate derivatives (BIS)

The LIBOR-to-SOFR Transition

One of the most consequential structural changes in money markets: the transition from USD LIBOR to SOFR as the primary short-term dollar benchmark.

LIBOR A panel-based, *unsecured* rate — an average of rates at which banks *reported* they could borrow; at its pre-transition peak the reference rate for ~\$350 trillion in contracts. Its fatal flaw emerged in 2012: manipulation allegations showed banks had submitted false quotes, rendering LIBOR untrustworthy. Panel LIBOR ceased on 30 June 2023; synthetic USD LIBOR ended 30 September 2024

SOFR Administered by the Federal Reserve Bank of New York; a *transaction-based, secured* rate — a volume-weighted median of actual overnight repo transactions collateralized by US Treasury securities

LIBOR vs. SOFR

Feature	USD LIBOR (ceased 2023)	SOFR
Rate type	Unsecured	Secured (repo collateral)
Determination	Panel survey	Transaction-based
Tenor	Was overnight–12m (ceased)	Overnight (Term SOFR derived)
Credit risk premium	Yes (bank credit)	No (near-zero, Treasury collateral)
Manipulation risk	High (demonstrated)	Low (volume-based median)

Money Market Yields: Special Quoting Conventions

Money market securities use special rate quoting conventions:

Discount yield (i_{dy}) Interest rate quoted on an annual basis, assuming a **360-day year**, as a percentage of **face value**:

$$i_{dy} = \frac{F - P_0}{F} \times \frac{360}{n}$$

Single-payment yield (i_{sp}) Interest rate quoted on an annual basis, assuming a **360-day year**, as a percentage of **purchase price**

Both may be converted to a **bond equivalent yield (i_{bey})** for comparison with bonds.

The Bond Equivalent Yield

$$i_{bey} = \frac{F - P_0}{P_0} \times \frac{365}{n}$$

Two differences from the discount yield: the denominator is P_0 (purchase price) rather than face value, and the calculation uses **365** rather than 360 days.

- The discount yield is the *quoted* rate for T-bills; the BEY is the true comparable yield for investment decisions
- The BEY will always be **higher** than the discount yield for the same T-bill — it more accurately reflects the true annualized rate of return

Single-Payment Yields and Conversion

Negotiable CDs and federal funds pay interest only at maturity — quoted using **single-payment yields**:

$$i_{sp} = \frac{F - P_0}{P_0} \times \frac{360}{n} \quad \implies \quad i_{bey} = i_{sp} \times \frac{365}{360}$$

Converting to a bond equivalent yield allows fair comparison of different money market instruments.

Comparing Money Market Yields: Example

A \$1M investment in 90-day commercial paper has a 2% discount yield; an equivalent size and risk 90-day CD has a 2% single-payment yield. **Which security offers the better return?**

① **Commercial paper** ($i_{dy} = 2\%$):

$$P_0 = \$1,000,000 \times \left(1 - 0.02 \times \frac{90}{360}\right) = \$995,000, \quad i_{bey} = \frac{5,000}{995,000} \times \frac{365}{90} = 2.038\%$$

② **CD** ($i_{sp} = 2\%$):

$$F = \$1,000,000 \times \left(1 + 0.02 \times \frac{90}{360}\right) = \$1,005,000, \quad i_{bey} = \frac{5,000}{1,000,000} \times \frac{365}{90} = 2.028\%$$

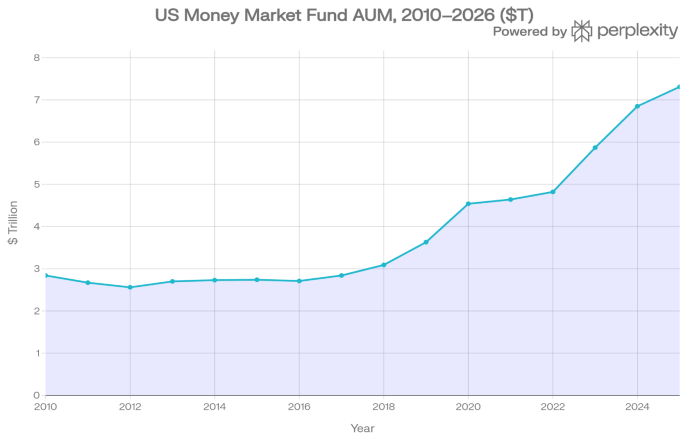
The commercial paper has the better return — its bond equivalent yield (2.038%) exceeds the CD's (2.028%).

Money Market Funds (MMFs)

MMFs are **pooled investment vehicles** that invest primarily in money market securities — T-bills, commercial paper, CDs, repos. They function as *de facto* bank accounts for institutional cash. Primary objectives:

- Preserve capital
- Provide daily liquidity
- Provide competitive yields to investors
- Maintain a stable net asset value (NAV), typically \$1 per share

Size and Growth of MMFs

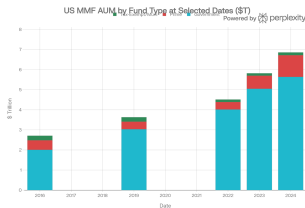


MMF Types and Structures

Government MMFs Invest $\geq 99.5\%$ in US government securities, repo collateralized by government securities, and cash; must maintain a stable \$1.00 NAV for all investor classes

Prime MMFs Invest in any money market instrument including CP, CDs, and repo; retail prime funds maintain stable NAV, institutional prime funds must use floating NAV

Tax-exempt/municipal MMFs Invest in short-term municipal paper; income is federally tax-free

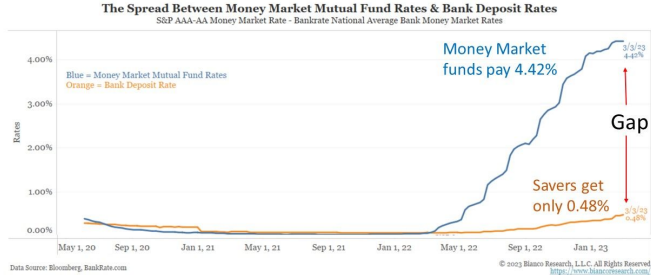


MMF Types: Government vs. Prime

Type	Primary investments	Key features	Typical investors
Government	U.S. Treasury securities, agency debt, and repos backed by government securities (at least 99.5% government assets for Treasury-only funds)	Lowest risk; exempt from certain liquidity fee rules; stable NAV; taxable income; yields tied to the Fed funds rate	Risk-averse investors seeking safety
Prime	Corporate commercial paper, bank CDs, and some government securities	Higher yields than government funds but slightly more credit risk; may include floating NAV for institutional class	Investors balancing yield and safety

Money Market Fund Rates vs. Bank Deposit Rates

Why would anyone want to put their money in a bank account?



MMF Regulation and Yields

SEC Rule 2a-7 (Investment Company Act of 1940) governs all US MMFs:

- Portfolio maturity limits: $WAM \leq 60$ days; $WAL \leq 120$ days
- Credit quality: only “eligible securities” rated A-1/P-1 or equivalent
- Diversification: no single issuer $> 5\%$ (except government securities)
- Liquidity minimums (effective April 2024): daily $\geq 25\%$; weekly $\geq 50\%$
- October 2024: mandatory liquidity fees on institutional prime/tax-exempt funds; discretionary redemption gates removed — prime funds now hold only $\sim 16\%$ of MMF assets (vs. $\sim 28\%$ pre-reform)
- Reporting: monthly Form N-MFP to the SEC, publicly available

MMF yields respond almost mechanically to the fed funds rate — portfolio maturities are so short (WAM typically 30–45 days) that most holdings reprice within weeks:

$$\text{MMF yield} \approx \text{EFFR} - \text{expenses} \pm \alpha$$

- Government MMFs track the Fed funds rate minus management fees ($\sim 10\text{--}15$ bps)
- Prime MMFs earn a spread above government MMFs ($\sim 10\text{--}30$ bps) by accepting CP and CD credit exposure

MMFs in China: Yu'e Bao

Yu'e Bao is a hugely popular money market fund offered through Alipay, managed primarily by Tianhong Asset Management.

- Launched 2013; lets users invest idle cash (“leftover” change from transactions) in a low-risk MMF
- The world's largest money market fund around 2017–2019; at its peak (Q1 2018) over ¥1.69 trillion (~\$260–268 billion) AUM, surpassing major US funds like JPMorgan's US Government Money Market Fund
- Growth driven by convenience, competitive yields, and Alipay's massive user base — then a long decline: by end-June 2026 AUM was down to ~¥680 billion (~\$94 billion), roughly 60% below peak, with the 7-day annualized yield below 1% since May 2026 (vs. 6–7% in its early years)
- From 2017–2018 Chinese regulators curbed risks: subscription caps for individuals (later relaxed), requirements for more liquid assets and reduced lower-quality exposure, plus competition from bank wealth products — it was overtaken by US institutional funds from JPMorgan and Fidelity; falling Chinese money-market yields then compressed both its size and its appeal

MMFs: What Are the Risks?

MMFs aim for low volatility and are among the safest investments — but not entirely risk-free:

Interest rate risk Yields fall with declining rates, potentially underperforming inflation

Credit risk Minimal but possible if issuers default

Liquidity risk In market stress, funds may face redemption pressures, leading to forced sales at losses

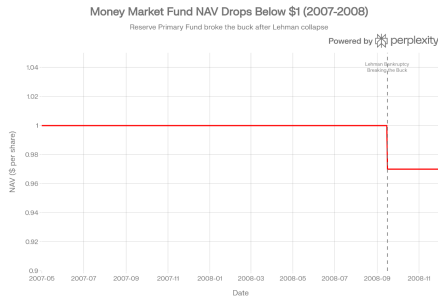
No government insurance Unlike bank deposits, no FDIC protection; SIPC covers only brokerage failures

Opportunity cost Lower returns than stocks or longer-term bonds

Mini Case: The 2008 GFC and the Reserve Primary Fund

The 2008 crisis highlighted MMF vulnerabilities when the Reserve Primary Fund “**broke the buck.**”

- After Lehman Brothers' bankruptcy on September 15, the fund faced a massive wave of redemptions (a “run”)
- On September 16 it wrote its Lehman holdings down to zero; NAV fell from the stable \$1.00 to **\$0.97** — only the second time in history a MMF broke the buck
- The fund suspended redemptions for orderly liquidation; investors eventually recovered ≈ 99.1 cents on the dollar, but the final distribution did not occur until **December 2014**, leaving capital frozen for years



Key Takeaways

- **Money market:** short-term (≤ 1 year), highly liquid, low-risk securities; purpose is managing short-term funding needs and investing excess cash safely, not long-term growth
- **Instruments:** T-bills (default-risk free, sold at discount); fed funds (overnight unsecured interbank loans; the Fed funds rate is the policy target); repos (collateralized by Treasuries; rate $<$ Fed funds); CP (unsecured corporate debt, limited secondary market); negotiable CDs (bank time deposits, tradable); BAs (bank-guaranteed trade finance, low risk)
- **Yields:** special quoting conventions (discount yield, single-payment yield); the bond equivalent yield standardizes yields for comparison with bonds
- **MMFs:** mutual funds investing in money market securities; goals — preserve capital, provide liquidity, maintain stable \$1.00 NAV. Not risk-free: “breaking the buck” in 2008, no FDIC insurance, interest rate and credit risk